

Irfan Sudarani

+91 6361850072 — irfanrs024@gmail.com — linkedin.com/in/irfanrs
github.com/Irfan-RS — codolio.com/profile/irfan72 — irfansudarani.netlify.app

Summary

Seeking a Software Development Intern role to contribute to building scalable and impactful systems, with a strong focus on problem-solving in Full-Stack Web Development. Experienced in developing APIs, working with databases, and building backend systems, with interest in system design and production-ready applications.

Education

B.M.S. College of Engineering, Bengaluru	2023 – 2027
<i>B.E. in Computer Science and Engineering</i>	CGPA: 8.57
Smt. Vidya P Hanchimani PU College, Dharwad	2021 – 2023
<i>Pre-University Course (PCMC)</i>	KCET Rank: 2674

Skills

Languages: Java, Python, C, JavaScript

Databases: MongoDB, Cassandra, MySQL, PostgreSQL

Core: Data Structures And Algorithm, DBMS, OOPs, Operating Systems, Computer Networks, System Design, Basic Cloud Technologies.

Tools: Git, GitHub, VS Code, Docker, Postman, Figma

OS: Windows, Ubuntu

Projects

Cognitive Interview AI [GitHub](#) *Technologies: React, FastAPI, Python, LLM (Ollama/OpenAI-compatible), RAG, ChromaDB, SQLAlchemy, SQLite, MediaPipe, WebRTC/MediaRecorder, faster-whisper, Google Cloud TTS, React Router, Tailwind CSS*

- Built a full-stack AI mock-interview platform using **React** and **FastAPI**, with a RAG-powered question engine over a **3000+ question bank**, using **ChromaDB** vector search and LLM-driven semantic and metadata matching to personalize questions by role, resume, or topic.
- Engineered a real-time computer vision pipeline using **MediaPipe** face landmarks for in-browser gaze and eye-contact tracking with custom **5-point calibration**, feeding live engagement signals into an LLM-based answer analysis system for grammar, relevance, and filler-word detection.
- Designed an adaptive interview engine that dynamically adjusts question difficulty and generates context-aware LLM follow-up questions based on candidate responses, with asynchronous **FastAPI** services for STT/TTS pipelines and LLM-synthesized performance analytics.

EcoLoop Building Agent [GitHub](#) *Technologies: Python, FastAPI, React, Vite, Tailwind CSS, PyEnergyPlus C API, EnergyPlus, SQLite*

- Developed an AI-driven building energy optimization platform integrating **EnergyPlus** simulations with a **FastAPI** backend to evaluate HVAC strategies and optimize building energy consumption.
- Built a React-based interactive dashboard for configuring building parameters, running simulations, and visualizing energy and HVAC performance metrics.
- Designed an automated optimization workflow that analyzes simulation outputs and recommends energy-efficient HVAC configurations while balancing occupant comfort and operational constraints.

Campus Space [GitHub](#) [Live](#) *Technologies: React.js, Tailwind CSS, Node.js, Express.js, MongoDB (Atlas), AWS S3, Firebase Auth*

- Built a responsive campus community platform with an intuitive user interface for creating posts, interacting through likes and comments, and managing user profiles.
- Refactored the backend into a modular layered architecture with clear separation of configuration, business logic, data models, routes, and middleware, reducing code coupling by 80% and improving maintainability.
- Implemented secure cloud-based media storage with automated cleanup of orphaned profile images, preventing unnecessary storage usage and reducing hosting costs.
- Developed a personalized dashboard and paginated notification system for likes, comments, and threaded replies, with optimized database queries enabling sub-second response times.

QuickCare – Healthcare UI/UX Design [Figma](#)

- Designed a user-friendly healthcare interface focused on intuitive navigation, accessible layouts, and a seamless patient experience.
- Created interactive prototypes covering core healthcare workflows with consistent visual hierarchy, responsive layouts, and clear user interactions.

Leadership

Senior Coordinator – NSS Unit, BMSCE

Oct 2023 – Present

- Represented VTU and Karnataka at the National Integration Camp (NIC) at the national level.
- Planned and executed a 7-day residential NSS camp at a government school, managing logistics, scheduling, and on-ground coordination for two consecutive years.
- Organized and coordinated 50+ NSS events ensuring smooth execution and active student participation.